

Monitor Resources with **Azure Monitor**

Metrics, logs, alerts, diagnostic settings, Log Analytics, Application Insights, and Azure Advisor — how Azure Monitor works, what it collects, and how to turn data into actionable alerts.

How Azure Monitor fits together

Azure Monitor is the umbrella — everything Azure collects flows through it. Data sources send either Metrics (numeric time-series, stored 93 days, near real-time) or Logs (structured records, queried with KQL, stored in Log Analytics workspaces). Alerts sit on top of both and trigger action groups when thresholds are crossed.

AZURE MONITOR DATA FLOW

SOURCES

Azure Resources

VMs, storage, networking, app services — emit platform metrics automatically

SOURCES

Guest OS / Apps

Requires Azure Monitor Agent or App Insights SDK to collect OS metrics and app telemetry

SOURCES

Activity Log

Control-plane events — who did what to which resource, subscription-level

↓ all data flows into Azure Monitor

STORE

Metrics

Numeric time-series, 93 day retention, queryable in Metrics Explorer

STORE

Log Analytics Workspace

Structured log data, KQL queries, configurable retention (30–730 days)

↓ alerts sit on top of both stores

ACTION

Alert Rules

Metric alerts, log search alerts, activity log alerts

ACTION

Action Groups

Email, SMS, push, webhook, Logic App, Function, ITSM, Automation runbook

ACTION

Dashboards / Workbooks

Visualize metrics and log data for teams

Eight terms you need cold

Metrics

Numeric time-series data collected from Azure resources automatically — CPU %, network bytes, disk IOPS. Stored for 93 days. Viewable in Metrics Explorer without any configuration.

Tip: Platform metrics are free and require no agent. Guest OS metrics (memory, disk space) require the Azure Monitor Agent.

Diagnostic Settings

The configuration that enables a resource to send its platform logs and metrics to a destination: Log Analytics workspace, storage account, Event Hub, or partner solution.

Tip: Without a diagnostic setting, resource logs are NOT collected — they must be explicitly enabled per resource.

Log Analytics Workspace

The central store for all log data in Azure Monitor. KQL (Kusto Query Language) is used to query and analyze the data. Multiple resources and subscriptions can send logs to one workspace.

Tip: Keep workspaces to a minimum — one per environment is usually enough. Fewer workspaces = easier cross-resource querying.

Alert Rule

A condition + threshold that triggers an action. Three types: Metric alert (numeric threshold), Log search alert (KQL query result), Activity log alert (control plane event).

Tip: Metric alerts fire faster (1–5 min) than log search alerts (minimum 5 min evaluation window). Use metrics for real-time alerting on resource health.

Action Group

A reusable set of notification and automation actions triggered when an alert fires: email, SMS, push, voice call, webhook, Logic App, Azure Function, ITSM, Automation runbook.

Tip: Separate alert rules from action groups — one action group can be reused across many alert rules, reducing maintenance.

Activity Log

A subscription-level audit log of all control-plane operations — who created, modified, or deleted a resource, when, from which IP. Retained for 90 days by default.

Tip: Send Activity Log to a Log Analytics workspace via Diagnostic Settings to extend retention beyond 90 days and enable KQL queries.

Application Insights

An Application Performance Management (APM) service that collects telemetry from web apps — request rates, response times, exceptions, dependency calls, and custom events.

Tip: Application Insights stores data in a Log Analytics workspace — query it with KQL alongside infrastructure logs for full-stack correlation.

Azure Advisor

A free recommendation engine that analyzes resource configuration and usage to provide personalized recommendations across five categories: Cost, Security, Reliability, Operational Excellence, Performance.

Tip: Advisor integrates with Azure Monitor alerts — you can set up alerts to fire when new high-severity recommendations appear.

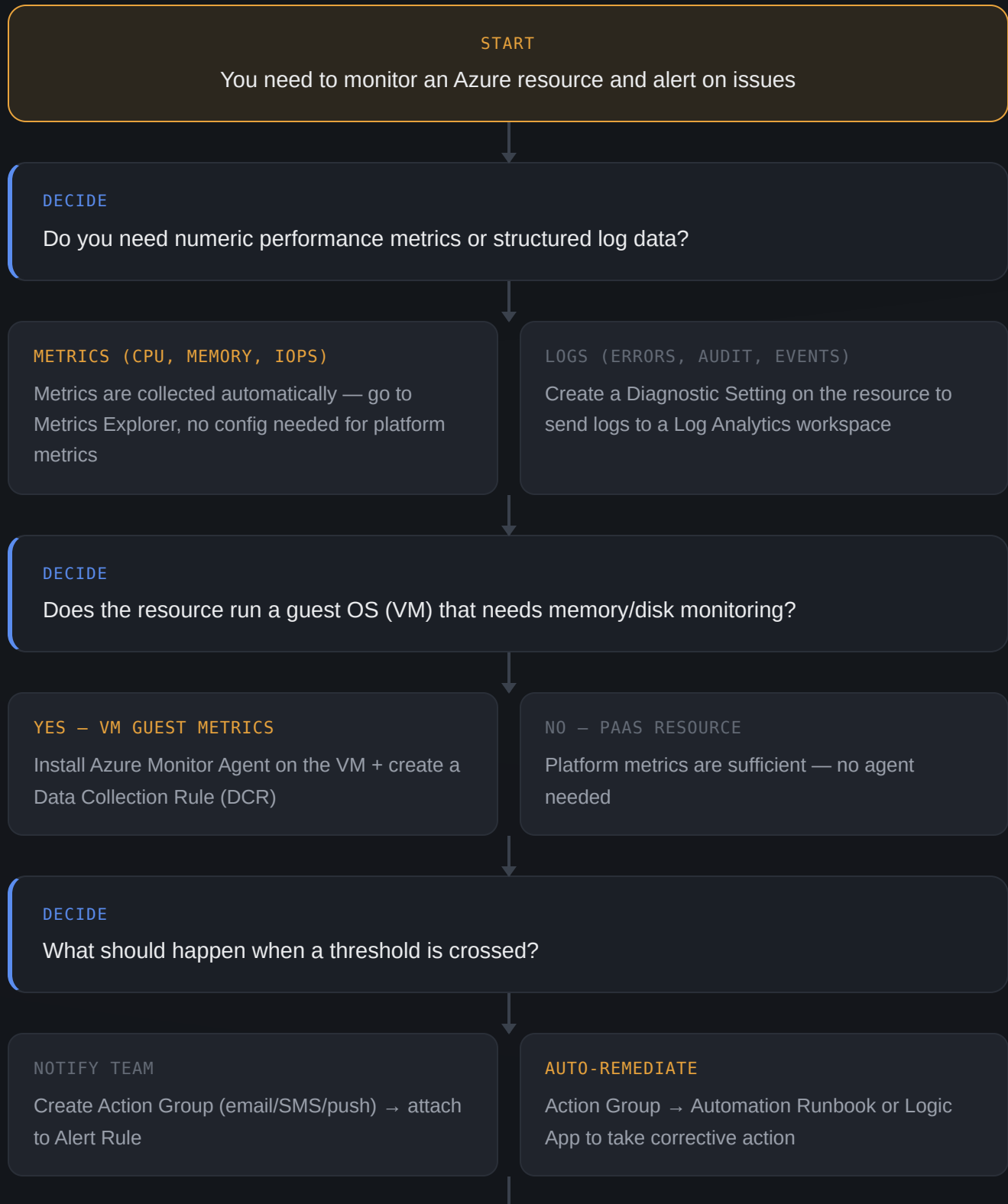
DATA TYPE	STORE	DEFAULT RETENTION	QUERY LANGUAGE	AGENT NEEDED?
Platform Metrics	Azure Monitor Metrics	93 days	Metrics Explorer / REST	No
Guest OS Metrics	Azure Monitor Metrics	93 days	Metrics Explorer	Yes — Azure Monitor Agent
Resource Logs	Log Analytics Workspace	30 days (configurable)	KQL	No — needs Diag Setting
Activity Log	Azure Monitor / LAW	90 days (extend via LAW)	KQL (in LAW)	No
App telemetry	App Insights / LAW	90 days (configurable)	KQL	Yes — SDK or auto-instrument

EXAM GOTCHA — RESOURCE LOGS ARE NOT AUTOMATIC

Platform **metrics** are collected automatically for every Azure resource. Platform **logs** (resource logs, formerly called diagnostic logs) are **not** — you must create a Diagnostic Setting on each resource to enable log collection and specify where to send them. Without it, those logs are silently dropped.

Setting up monitoring for a resource

Walk through this whenever you need to monitor a new resource or set up alerting on an existing one.



DECIDE

Alert on metrics or logs?

NEAR REAL-TIME (1-5 MIN)

Metric alert rule — fastest, no LAW needed

LOG-BASED (CUSTOM QUERY)

Log search alert rule — KQL query, minimum 5 min evaluation, requires LAW

CONTROL PLANE EVENTS

Activity log alert — fires on resource create/delete/modify events

VISUALIZE

Pin metrics to a shared Dashboard or build a Workbook in Azure Monitor for team visibility

DONE

Resource monitored, logs collected, alerts configured, team notified automatically

Tools for Azure Monitor

>_ Azure CLI — Diagnostic Settings

BEST FOR: enabling resource log collection at scale

Enable diagnostic settings across many resources in a script — critical first step before any log-based alerting or querying will work.

```
az monitor diagnostic-settings
create \
  --name "diag-vm-web01" \
  --resource $(az vm show -g rg-
prod \
  -n vm-web01 --query id -o tsv)
\
  --workspace $(az monitor log-
analytics workspace show \
  -g rg-prod -n law-prod --query
id -o tsv) \
  --logs
'[{ "category": "Administrative", "ena
bled": true}]' \
  --metrics
'[{ "category": "AllMetrics", "enabled
": true}]'
```

>_ Azure CLI — Alert Rules & Action Groups

BEST FOR: scripted alert setup

Create action groups and metric alert rules in one script — reuse the same action group across multiple alert rules.

```
# Create action group
az monitor action-group create \
  -g rg-prod -n ag-ops \
  --action email ops-team
ops@contoso.com

# Create metric alert (CPU > 90%)
az monitor metrics alert create \
  -g rg-prod -n alert-cpu-high \
  --scopes $(az vm show -g rg-prod
\
  -n vm-web01 --query id -o tsv)
\
  --condition "avg Percentage CPU >
90" \
  --window-size 5m \
  --evaluation-frequency 1m \
  --action ag-ops \
  --severity 2
```

KQL — Log Analytics queries

BEST FOR: querying logs, building log search alerts

KQL is the query language for all Log Analytics data. These patterns cover the most exam-relevant and day-to-day useful queries.

```
// VM heartbeat – check if VM is reporting
Heartbeat
| where TimeGenerated > ago(1h)
| summarize LastHeartbeat = max(TimeGenerated)
    by Computer
| where LastHeartbeat < ago(5m)

// Activity log – who deleted a resource?
AzureActivity
| where OperationNameValue endswith "delete"
| where ActivityStatusValue == "Success"
| project TimeGenerated, Caller, ResourceGroup, Resource, OperationNameValue

// Performance – avg CPU last hour
Perf
| where ObjectName == "Processor"
| where CounterName == "% Processor Time"
| summarize avg(CounterValue) by Computer, bin(TimeGenerated, 5m)
```

Azure CLI — Log Analytics workspace

BEST FOR: workspace creation and management

Create workspaces, set retention, and run KQL queries directly from the CLI — useful for pipeline automation and quick diagnostic queries.

```
az monitor log-analytics workspace create \ -g rg-prod -n law-prod \ --location eastus \ --retention-time 90
az monitor log-analytics workspace update \ -g rg-prod -n law-prod \ --retention-time 365
az monitor log-analytics query \ --workspace $(az monitor log-analytics workspace show \ -g rg-prod -n law-prod --query id -o tsv) \ --analytics-query "Heartbeat | summarize count() by Computer"
```


<> Bicep — Diagnostic Settings at scale

BEST FOR: IaC — enable monitoring at deploy time

Deploy diagnostic settings alongside every resource in Bicep — monitoring is never an afterthought, it's baked into the template.

```
resource diagSetting
'Microsoft.Insights/
diagnosticSettings@2021-05-01-
preview' = {
  name: 'diag-${vm.name}'
  scope: vm
  properties: {
    workspaceId: lawWorkspace.id
    metrics: [{
      category: 'AllMetrics'
      enabled: true
      retentionPolicy: { enabled:
true, days: 30 }
    }]
    logs: [{
      categoryGroup: 'allLogs'
      enabled: true
    }]
  }
}
```

ALERT TYPE	DATA SOURCE	MIN FREQUENCY	BEST FOR
Metric alert	Azure Monitor Metrics	1 minute	CPU, memory, disk — fast response
Log search alert	Log Analytics workspace	5 minutes	Custom KQL conditions, log-based thresholds
Activity log alert	Azure Activity Log	Near real-time	Resource create/delete/modify events
Service health alert	Azure Service Health	Real-time	Azure outages, planned maintenance, advisories

Principles worth internalizing

AZURE MONITOR AGENT VS LEGACY AGENTS

The **Azure Monitor Agent (AMA)** is the current unified agent for collecting guest OS metrics and logs — it replaces the older Log Analytics agent (MMA/OMS) and Diagnostics extension (WAD/LAD). AMA uses **Data Collection Rules (DCRs)** to define what data to collect and where to send it. The exam still references both — know that AMA + DCR is the modern approach and legacy agents are being deprecated.

EXAM GOTCHA – METRICS VS LOGS RETENTION

Platform **metrics** are retained for **93 days** and cannot be extended — export to a storage account or Log Analytics workspace to keep them longer. **Resource logs** in a Log Analytics workspace default to **30 days** but can be extended to 730 days (paid). The Activity Log defaults to **90 days** in the subscription — export it to a workspace to extend retention and enable KQL queries.

Six mistakes that show up on the exam

Expecting resource logs without a Diagnostic Setting

Platform metrics are automatic. Resource logs are not — every resource needs an explicit Diagnostic Setting pointing to a destination before logs are collected.

Expecting memory/disk metrics without an agent

CPU and network are platform metrics — no agent needed. Memory usage and disk space require the Azure Monitor Agent + a Data Collection Rule on the VM.

Using log search alerts when metric alerts are faster

Log search alerts have a 5-minute minimum evaluation window. Metric alerts can fire in 1 minute. For real-time CPU or memory alerting, always use metric alerts.

Creating one action group per alert rule

Action groups are reusable — create one per team or severity level and attach it to many alert rules. Duplicating the same action group adds maintenance overhead with no benefit.

Not extending Activity Log retention

The Activity Log only retains 90 days by default. For compliance or forensics, export it to a Log Analytics workspace to extend retention and gain KQL queryability.

Confusing Application Insights with Azure Monitor

Application Insights is a feature of Azure Monitor focused on app telemetry (requests, exceptions, dependencies). It stores data in a Log Analytics workspace — both are part of Azure Monitor, not separate products.